

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 June 2026
Team ID	SWTID-2026-5525
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story/Sub-Task)
FR-1	Data Collection and Storage	Collect EV datasets, Import CSV files, Store data in MySQL database
FR-2	Data Processing and Analysis	Perform SQL operations, Clean datasets, Generate analytical queries
FR-3	EV Data Visualization	Create Tableau dashboards, KPI cards, Charts, Maps and Filters
FR-4	Web Dashboard Integration	Publish Tableau dashboard, Embed dashboard using Flask web application

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Application should provide a simple and user-friendly interface to analyze EV insights
NFR-2	Security	Data access and dashboard publishing should be managed securely
NFR-3	Reliability	Dashboard should provide accurate visualization from processed EV datasets
NFR-4	Performance	System should load Tableau dashboards efficiently with large datasets
NFR-5	Availability	Published dashboard should be accessible through Tableau Public and web application
NFR-6	Scalability	System should support additional EV datasets and future enhancements